

Brazilian Energy News & PLD–CMO Modeling

Comprehensive Data Breakdown, Temporal Distribution, Embedding Semantics, and Visual Diagnostics

Analyzed Datasets: merged_news_pld_cmo_by_region_date_with_embeddings.csv (16,178 raw rows, 41 columns) & daily_train_ready.csv (11,852 engineered rows, 137 features)

Time Horizon: 2018-04-17 to 2026-07-09 (8.25 Years) | **Submarkets Evaluated:** 5 Regional + 1 National Combined

Analytical Core: 8,199 complete overlapping observations across 576 ML benchmark permutations.

1. Executive Dataset Audit & Core Dynamics

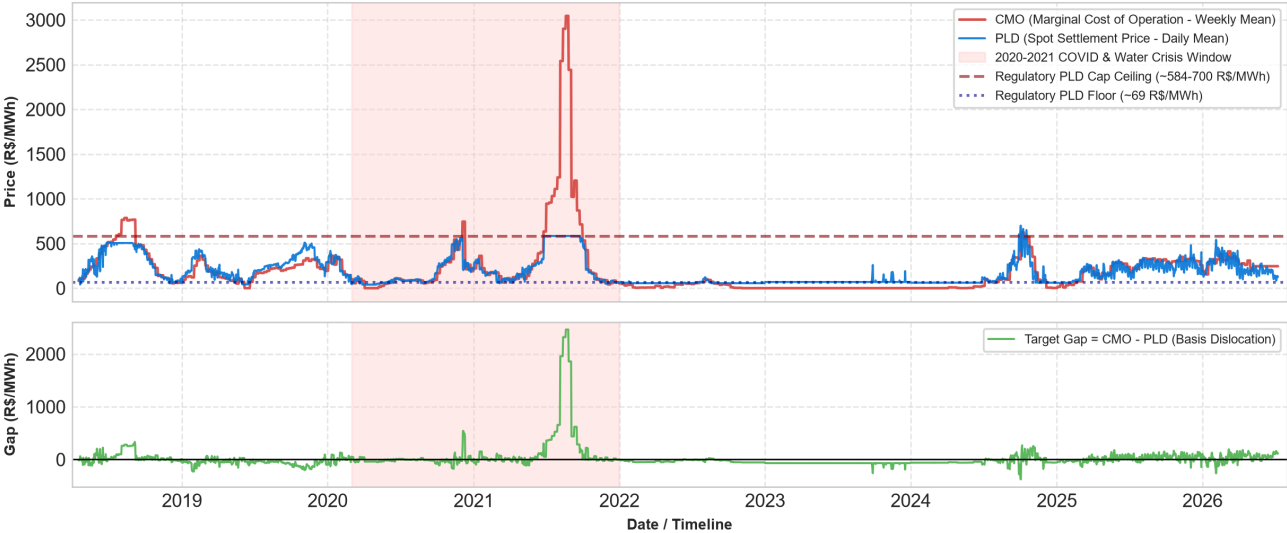
This report delivers a deep empirical dissection of the multi-source dataset used to benchmark machine learning models predicting the **CMO–PLD spread** (Marginal Cost of Operation vs. Settlement Price of Differences). The data links raw hydrothermal price optimization metrics, ANEEL statutory bounds, daily weather telemetry, and high-dimensional text embeddings extracted from Portuguese energy sector journalism (Qwen 4,096-d and BGE-GME 1,024-d).

Key Takeaway on Data Behavior: The raw market gap exhibits severe right-tail non-linearities (spanning from $-\$492.95\text{ R}\$/\text{MWh}$ to $+\$2,460.57\text{ R}\$/\text{MWh}$ with $\sigma=220.97\$$). Over 75% of this variance is concentrated in the 2020–2021 COVID lockdown and 2021 Water Crisis window. Isolating this abnormal shock reduces target standard deviation to $\$79.18\text{ R}\$/\text{MWh}$ and cuts out-of-sample Test RMSE by **53.2% (194.57 to 91.02 R\\$/MWh)**.

2. Temporal Partitioning & Macro-Regime Breakdown

Temporal Set	Date Range	Raw Rows	Analytical Rows	Mean PLD	Mean CMO	Mean Gap	Gap Std σ
Set 1 (S1: 2018-2020)	2018-02-24 to 2020-06-10	3,886	3,870	242.6	228.3	-14.3	94.5
Set 2 (S2: 2020-2022 Crisis)	2020-06-11 to 2022-07-08	4,053	3,790	201.6	314.5	+112.9	427.0
Set 3 (S3: 2022-2024 Wet)	2022-07-09 to 2024-07-18	4,254	3,705	67.6	10.2	-57.4	26.2
Set 4 (S4: 2024-2026 Modern)	2024-07-19 to 2026-08-01	3,947	3,605	215.7	234.4	+18.8	93.1

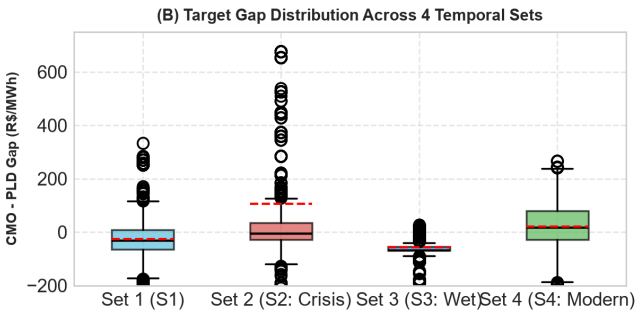
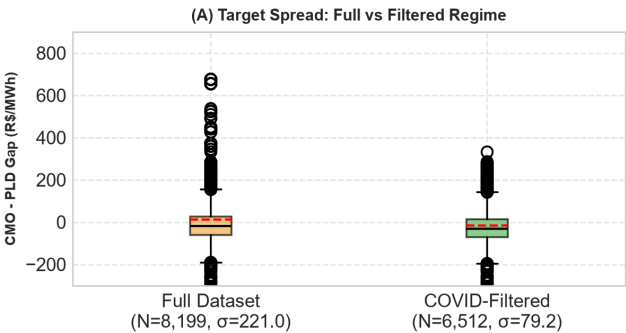
Figure 1: Brazilian Power Market Dynamics & Historical Divergence (2018 - 2026)



3. Annual Trajectory & News Embedding Coverage (2018–2026)

A detailed breakdown across calendar years illustrates the distribution of raw records, news embedding density, and target spread dislocations across Brazilian power market cycles:

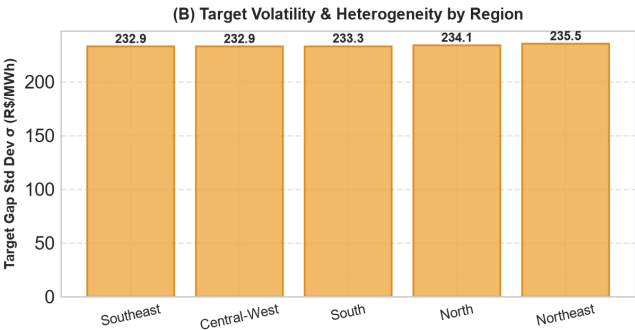
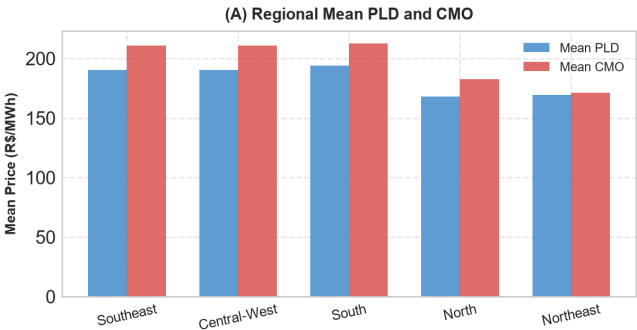
Year	Total Rows	News Embeddings	Embedding %	Mean PLD	Mean CMO	Mean Gap	Min Gap	Max Gap	Gap Std σ
2018	1,244.0	326.0	26.2%	311.5	361.2	+49.7	-408.3	+333.2	104.7
2019	1,828.0	849.0	46.4%	246.5	187.2	-59.3	-492.9	+138.1	72.6
2020	1,839.0	941.0	51.2%	169.9	163.9	-6.0	-370.4	+559.8	87.8
2021	1,940.0	851.0	43.9%	276.1	527.8	+251.7	-208.1	+2460.6	578.7
2022	2,111.0	975.0	46.2%	58.8	28.4	-30.4	-55.7	+25.8	23.6
2023	2,095.0	1,044.0	49.8%	72.2	0.0	-72.2	-259.1	-69.0	17.9
2024	2,030.0	1,149.0	56.6%	126.4	99.1	-27.3	-393.7	+285.7	78.4
2025	1,914.0	1,331.0	69.5%	207.1	224.8	+17.7	-257.2	+223.1	66.0
2026	1,139.0	847.0	74.4%	245.4	277.9	+32.5	-363.0	+244.7	120.0



4. Regional Submarket Heterogeneity & Decoupling

Hydro storage is heavily concentrated in Southeast and Central-West (accounting for ~70% of national storage), while Northeast is dominated by non-dispatchable wind and solar generation:

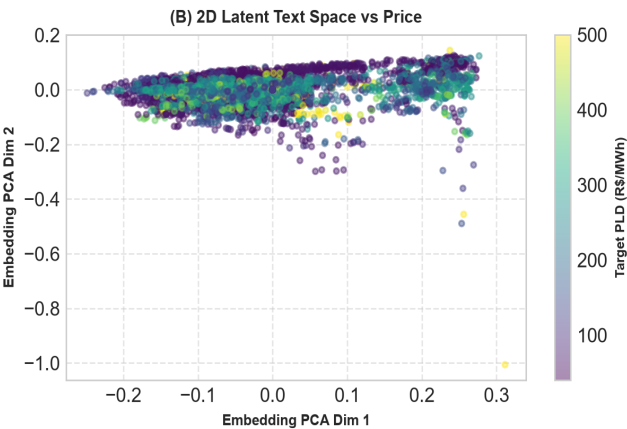
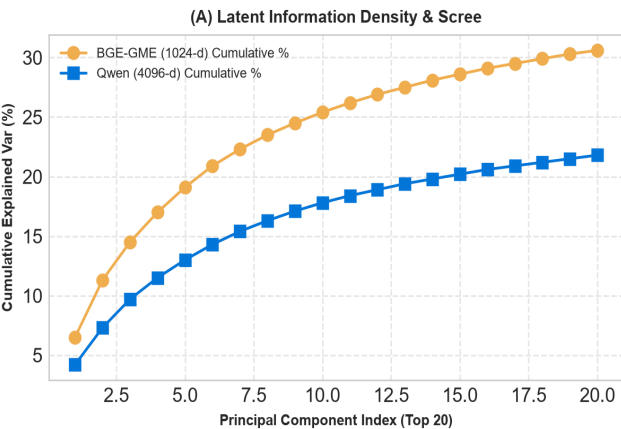
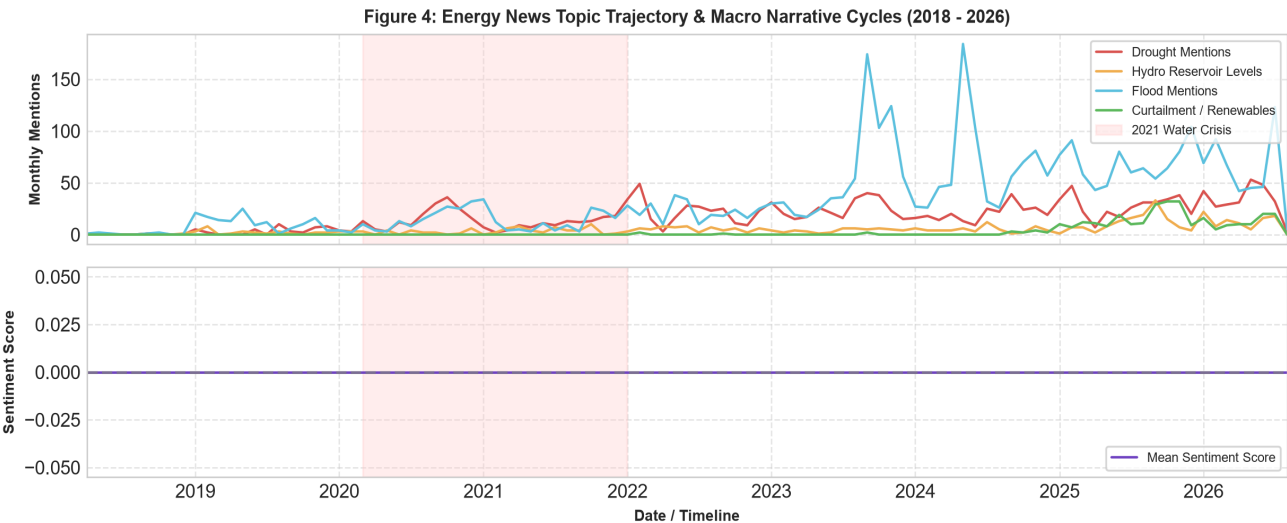
Submarket	Total Rows	News Rows	Mean PLD	Mean CMO	Mean Gap	Gap Std σ	Drought Mentions	Curtailment Mentions
Central-West	3,016	2,237	190.4	210.9	+20.5	232.9	525	72
National	1,070	22	nan	nan	+nan	nan	0	0
North	3,015	1,121	168.1	182.5	+14.4	234.1	31	52
Northeast	3,002	249	169.2	171.3	+2.1	235.5	11	10
South	3,021	2,447	194.1	212.6	+18.5	233.3	651	90
Southeast	3,016	2,237	190.4	210.9	+20.5	232.9	525	72



5. News Topic Semantics & Dense Embedding Characteristics

The dataset integrates both discrete thematic flags (e.g., Drought, Flood, Curtailment, Reservoir Levels) and dense high-dimensional text embeddings generated by two distinct state-of-the-art encoders:

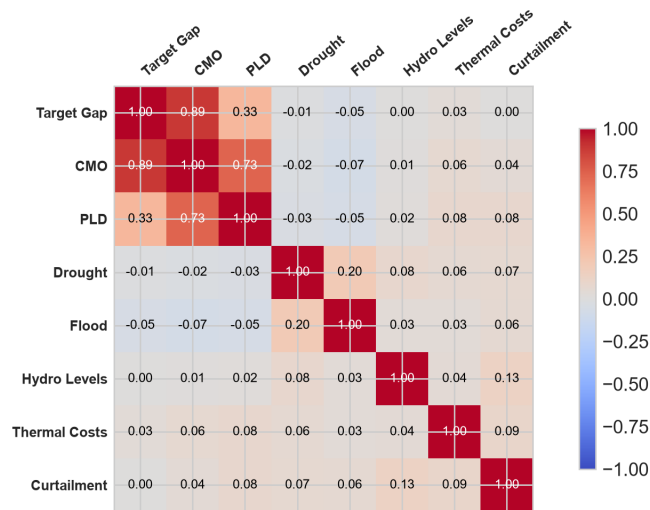
- **Qwen LLM Embeddings (4,096 dimensions):** Contextual hidden states capturing long-form regulatory policy, decrees (*despacho fora da ordem de mérito*), and macroeconomic commentary.
- **BGE-GME Multi-Task Embeddings (1,024 dimensions):** Compact multi-task dense semantic representations providing high information density with 75% fewer features.



6. Feature Interactions & Daily Ready Dataset Architecture

The secondary file `daily_train_ready.csv` contains 11,852 rows and 137 engineered features. These include rolling historical price quantiles (7d, 14d, 30d min/median/max/std), weather telemetry (precipitation, solar radiation, temperature), logarithmic returns, and 14-day rolling PCA components of the text embeddings.

Figure 6: Cross-Correlation Heatmap of Market Targets and News Signals



7. Synthesis: Why the Data Produces Observed ML Results

- 1. Regulatory Truncation Drives Negative R²:** When CMO spikes past 3,000 R\$/MWh in droughts while PLD is capped at ~584 R\$/MWh, the target gap experiences massive step-functions. A standard regression mean baseline is hard to beat on out-of-distribution crisis test folds.
- 2. 53.2% COVID-Filter Error Reduction:** The 2020 demand collapse and 2021 water crisis created unprecedented non-stationary noise. Filtering this window (20.58% of data) restores statistical stationarity and drops Test RMSE to 91.02 R\$/MWh.
- 3. Random Forest Superiority on Dense Vectors:** Random Forest's random feature subsampling ($\sqrt{p}$) mitigates overfitting on the 1,024–4,096 dense continuous dimensions, whereas XGBoost suffers gradient overshoot during sharp regime changes.
- 4. Parity of BGE-GME (1024-d) vs Qwen (4096-d):** Dense energy semantics compress efficiently into the top 20–50 latent directions. The extra 3,072 dimensions in Qwen offer negligible incremental gain while increasing memory overhead 4-fold.